

Diagnostic information - amplifier

Front left/right loudspeaker - mid-range/woofer - acoustic error

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on
- Battery positive voltage **10.8 V - 16 V**

Fault setting condition(s)

- Loudspeaker self-test not OK

Possible fault cause(s)

- Line faulty
- Lines and/or connectors corroded
- Loudspeaker diaphragm faulty
- Internal fault in loudspeaker
- Implausible signal

Fault effect(s)

- Loudspeaker has limited functionality or does not work
- Changed sound

Setpoint values for line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V ⇒ + 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **□ 0.9 Mohm**

Open circuit:

- ♦ Setpoint resistance **< 2 ohm**

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Front loudspeaker - polarity reversed

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Rear left/right loudspeaker - tweeter/mid-range - acoustic error

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00060A

Diagnostic information - amplifier

Amplifier - function restriction

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Amplifier detects internal faults

Possible fault cause(s)

- Software incompatible
- Amplifier faulty

Fault effect(s)

- Various function restrictions

Diagnostic information - amplifier

Amplifier - function restriction - temperature exceeded

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Temperature > 85 °C (> 185 °F)

Possible fault cause(s)

- Amplifier temperature too high
- Output stage temperature too high

Fault effect(s)

- Various function restrictions
- Reduced output-stage power output

Diagnostic information - amplifier

Amplifier - function restriction - temperature exceeded

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Temperature > 85 °C (> 185 °F)

Possible fault cause(s)

- Amplifier temperature too high
- Output stage temperature too high

Fault effect(s)

- Various function restrictions
- Reduced output-stage power output

00060D

Diagnostic information - amplifier

Amplifier - function restriction - temperature not reached

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Temperature < - 40 °C (- 40 °F)

Possible fault cause(s)

- Amplifier temperature too low

Fault effect(s)

- Various function restrictions

00061A

Diagnostic information - amplifier

Microphone - short circuit to ground

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Voltage < **1.2 V**

Possible fault cause(s)

- Short circuit to ground
- Lines and/or connectors corroded
- Microphone faulty

Fault effect(s)

- Microphone has limited functionality or does not work
- Telephone, cannot hear anything on other phone

Setpoint values for line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V** $\Rightarrow$ **+ 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance $\square$ **0.9 Mohm**

Open circuit:

- ♦ Setpoint resistance < **2 ohm**

Diagnostic information - amplifier

Microphone - short circuit to B+

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Voltage > **9 V**

Possible fault cause(s)

- Short circuit to B+
- Lines and/or connectors corroded
- Microphone faulty

Fault effect(s)

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Short circuit to ground:

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Open circuit:

- ♦ Setpoint resistance **< 2 ohm**

Diagnostic information - amplifier

Microphone - open circuit

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Current < **1 mA**

Possible fault cause(s)

- Open circuit
- Lines and/or connectors corroded
- Microphone faulty

Fault effect(s)

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00063B

Diagnostic information - amplifier

Amplifier - coding

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Amplifier reports incorrect coding

Possible fault cause(s)

- Coding incorrect or not carried out

Fault effect(s)

- Various function restrictions

Diagnostic information - amplifier

Authentication failed - signature/certificate

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Data transmission of the MOST components is encrypted using a protocol. The encryption cannot be verified by the communication partner.

Possible fault cause(s)

- The signature of the MOST control unit cannot be verified
- The communication certificate is not correct

Fault effect(s)

- No or restricted function of the relevant MOST component

Diagnostic information - amplifier

MOST (Media Oriented Systems Transport) bus - function restriction

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- No message/implausible message
- Fault value received

Possible fault cause(s)

- Messages are not transmitted or are not transmitted fully
- Fault memory entries for one or more MOST components prevent complete and plausible data transmission
- Messages received/sent by one or more MOST components are implausible
- Received/sent messages contain fault values

Fault effect(s)

- Amplifier has limited functionality or does not work
- PCM has limited functionality or does not work
- TV tuner has limited functionality or does not work
- Extended environment data in fault memory entries invalid

Diagnostic information - amplifier

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- TV tuner has limited functionality or does not work
- Extended environment data in fault memory entries invalid

B106C96

Diagnostic information - amplifier

Subwoofer - function restriction

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on
- Battery positive voltage **10.8 V - 16 V**

Fault setting condition(s)

- Subwoofer detects internal fault

Possible fault cause(s)

- Temperature > **90 °C (194 °F)**
- Voltage too high
- Voltage too low
- Current too high
- Subwoofer faulty

Test condition(s)

- Subwoofer is activated

Fault effect(s)

- Subwoofer has limited functionality or does not work
- Changed sound

B106CF3

Diagnostic information - amplifier

Subwoofer - acoustic error

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on
- Battery positive voltage **10.8 V - 16 V**

Fault setting condition(s)

- Subwoofer self-test not OK

Possible fault cause(s)

- Line faulty
- Lines and/or connectors corroded
- Internal fault in subwoofer
- Implausible signal

Fault effect(s)

- Subwoofer has limited functionality or does not work
- Changed sound

Setpoint values for line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V ⇒ + 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **□ 0.9 Mohm**

Open circuit:

- ♦ Setpoint resistance **< 2 ohm**

B126B01

Diagnostic information - amplifier

Amplifier - fan, faulty

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on
- Fan on > 2 minutes

Fault setting condition(s)

- Amplifier detects an electrical fault in the internal fan

Possible fault cause(s)

- Overtemperature protection active
- Open circuit in wiring
- Short circuit to ground/B+
- Fan faulty

Fault effect(s)

- Housing fan does not work or has limited function

B129B11

Diagnostic information - amplifier

Microphone - short circuit to ground

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Voltage < **1.2 V**

Possible fault cause(s)

- Short circuit to ground
- Lines and/or connectors corroded
- Microphone faulty

Fault effect(s)

- Microphone has limited functionality or does not work
- Telephone, cannot hear anything on other phone

Setpoint values for line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V** ⇒ **+ 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance □ **0.9 Mohm**

Open circuit:

- ♦ Setpoint resistance < **2 ohm**

B129B12

Diagnostic information - amplifier

Microphone - short circuit to B+

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Voltage > 9 V

Possible fault cause(s)

- Short circuit to B+
- Lines and/or connectors corroded
- Microphone faulty

Fault effect(s)

- Microphone has limited functionality or does not work
- Telephone, cannot hear anything on other phone

Setpoint values for line test

Short circuit to B+:

- ♦ Setpoint voltage - 0.5 V $\Rightarrow$ + 0.5 V

Short circuit to ground:

- ♦ Setpoint resistance $\square$ 0.9 Mohm

Open circuit:

- ♦ Setpoint resistance < 2 ohm

B129B13

Diagnostic information - amplifier

Microphone - open circuit

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Current < **1 mA**

Possible fault cause(s)

- Open circuit
- Lines and/or connectors corroded
- Microphone faulty

Fault effect(s)

- Microphone has limited functionality or does not work
- Telephone, cannot hear anything on other phone

Setpoint values for line test

Short circuit to B+:

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Short circuit to ground:

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Open circuit:

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B129C11

Diagnostic information - amplifier

Microphone - short circuit to ground

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- Ignition on
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Fault setting condition(s)

- Voltage < **1.2 V**

Possible fault cause(s)

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- Microphone faulty

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Setpoint values for line test

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Open circuit:

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B129C12

Diagnostic information - amplifier

Microphone - short circuit to B+

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Voltage > **9 V**

Possible fault cause(s)

- Short circuit to B+
- Lines and/or connectors corroded
- Microphone faulty

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Short circuit to ground:

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Open circuit:

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B129C13

Diagnostic information - amplifier

Microphone - open circuit

Diagnostic conditions

- Ignition on
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- Amplifier on

Fault setting condition(s)

- Current < 1 mA

Possible fault cause(s)

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Open circuit:

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B13A001

Diagnostic information - amplifier

Front left/right loudspeaker - woofer - electrical fault

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on
- Battery positive voltage **10.8 V - 16 V**

Fault setting condition(s)

- Open circuit: resistance **> 20 ohm**
- Short circuit to ground/B+: resistance **< 3 ohm**
- Short circuit between lines: resistance **< 0.5 ohm**

Possible fault cause(s)

- Open circuit
- Short circuit to ground/B+ in lines
- Short circuit between lines
- Lines and/or connectors corroded
- Loudspeaker diaphragm faulty
- Internal fault in loudspeaker
- Implausible signal

Fault effect(s)

- Loudspeaker does not work
- Changed sound

Setpoint values for line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V ⇒ + 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **□ 0.9 Mohm**

Open circuit:

- ♦ Setpoint resistance **< 2 ohm**